

Kirtan Patel

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SUMMARY

Graduate Computer Science student at UIC with three-plus years training and mentoring technical staff in a university IT environment, authoring troubleshooting documentation, and explaining technical concepts to faculty, staff, and students at every skill level. Comfortable across Python, Java, JavaScript, and SQL; project work spans security tooling, full-stack web, and open-source contribution. Open to roles in software engineering, security, and IT operations.

EDUCATION

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| University of Illinois Chicago (UIC)
<i>Master of Science in Computer Science</i> | Chicago, IL
<i>December 2026</i> |
| – Relevant Coursework: Network Security, Cryptography, Computer System Security, Wearable Technology, Data Visualization & Analytics | |
| University of Illinois Chicago (UIC)
<i>Bachelor of Science in Computer Science – Software Engineering Concentration; Minor in Mathematics</i> | Chicago, IL
<i>May 2025</i> |
| – Dean's List: College of Engineering – Fall 2023 & Fall 2024 | |
| – Relevant Coursework: Building Secure Computer Systems, Artificial Intelligence, Database Systems, Software Engineering 1 & 2, UI/UX, Software Design, Statistical Methods, Data Structures | |

SKILLS

Languages: Python, Java, JavaScript, SQL
Python: CLI tooling, socket and multi-process programming, REST API integration, local LLM orchestration (Ollama), scikit-learn
Frameworks & Libraries: React, Node.js, Express.js, Next.js, JavaFX
Databases: MongoDB, MySQL, PostgreSQL
IT Support & Infrastructure: TeamDynamix (TDX) ticketing, tier-2 escalation, classroom networking, AV and lecture-capture systems, hardware diagnostics and repair, printer fleet management
Developer Tools: Git/GitHub, Docker, Linux, Postman, Vercel
Methodologies: Agile (sprints, standups, iterative development), version control workflows, technical documentation, mentoring and pair-troubleshooting

WORK EXPERIENCE

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|---|---|
| Technology Solutions, Office of the Vice Chancellor for Innovation – UIC
<i>IT Helpdesk Support Manager</i> | Chicago, IL
<i>Dec 2024 – Present</i> |
| – Lead tier-2 helpdesk support in TeamDynamix (TDX) for escalated tickets covering classroom networking, AV systems, and campus-wide technology issues; coach junior specialists through resolution steps as escalations come in. | |
| – Authored troubleshooting documentation adopted across the support team, shortening new-hire ramp time. | |
| – Interview support-team candidates, weighing technical depth alongside ability to teach non-technical users. | |
| – Diagnose and repair laptops and client devices; guide users through both immediate fixes and root causes. | |
| – Own printer operations across East Campus end-to-end, including parts replacement, repairs, and paper refills. | |
| Technology Solutions, Office of the Vice Chancellor for Innovation – UIC
<i>Classroom Technology Support Specialist</i> | Chicago, IL
<i>Aug 2022 – Dec 2024</i> |
| – Trained and mentored incoming support specialists from day one through full independence, using a mix of shadowing, hands-on practice, and structured feedback on real tickets. | |
| – Supported professors with classroom equipment including projectors, audio systems, and lecture capture setups. | |
| – Deployed and configured computer labs for courses and campus events, including installs and access. | |
| – Provided printer support and maintenance across assigned campus locations. | |

PROJECTS

Red_Co-Author | *Python, Ollama, Streamlit, scikit-learn, Laminar*

In Progress

- Engineered a fully-local Python red-team and defensive monitor stack for the Co-Authoring Jailbreak (CoJP) LLM vulnerability, automating a two-armed attack pipeline (direct prompt vs CoJP-framed prompt) across multiple local target models orchestrated through Ollama and scoring both responses with a separate judge model on the HarDBench 1-to-5 harmfulness rubric.
- Trained a defensive monitor on the resulting attack dataset using local nomic-embed-text embeddings plus a scikit-learn logistic regression classifier; evaluated for generalization on held-out AdvBench (520 prompts) and HarDBench (400 prompts) splits and benchmarked head-to-head against a zero-shot llama3 baseline. Surfaced the full pipeline through a Streamlit dashboard with ASR-by-target heatmaps and ROC curves.

Mini Tor Anonymity Network | *Python, TCP Sockets, Curve25519, AES-256-GCM, HKDF, SHA-256*

- Implemented Tor’s three-hop onion routing protocol from scratch in Python as six independent processes (directory, guard/middle/exit relays, client, destination) communicating over real TCP, with telescoping EXTEND cells so the client only ever connects to the guard while middle and exit are reached through the growing circuit.
- Engineered the entire cryptographic stack in pure Python with no third-party crypto libraries: SHA-256 (FIPS 180-4), HMAC, HKDF (RFC 5869), X25519 Diffie-Hellman via Montgomery ladder (RFC 7748), AES-256 (FIPS 197), and AES-256-GCM (NIST SP 800-38D), each verified against published NIST and IETF test vectors.

DiffFence | *Python, OSV.dev, Ollama*

- Developed a local-first Python CLI that acts as a diff-aware security gate, scanning code changes for security vulnerabilities and production risks before a push or merge.
- Implemented baseline-on-first-run logic so subsequent scans surface only new findings, eliminating repeated noise and keeping developer focus on incremental risk introduced by each change.

Email Summarizer | *Python, Ollama, Telegram API*

- Created a Python tool that fetches unread emails, summarizes content using a locally-run Mistral model via Ollama, and assigns priority levels to each message.
- Delivers prioritized summaries directly to the user through a Telegram bot, reducing inbox overhead without relying on cloud-based LLM services.

KeyTracker | *Next.js, MongoDB, Vercel*

- Built and deployed a barcode-based key checkout system for UIC Technology Solutions, enabling staff to check keys in and out with real-time availability tracking.
- Implemented audit logging, conflict detection (409 responses for double-checkouts), and mismatch flagging to enforce access control integrity and maintain a reliable activity trail.

WeatherGenie | *ReactJs, OpenWeather API, Arduino*

- Engineered an interactive weather forecasting platform integrating the OpenWeather API to deliver real-time, location-aware weather updates via a responsive landing page.
- Combined Arduino-powered local sensor data with API-based global forecasts to deliver room-level precision for temperature, humidity, and environmental analytics with dynamic visualizations.

OPEN SOURCE

Curio (urban-toolkit/curio) | *Python, Flask, Socket.IO, React, TypeScript*

Credited Contributor

- Designed and built a real-time, multi-device collaboration layer for Curio, a dataflow framework for urban visual analytics, that lets multiple users edit the same workflow together over a shared network; presented the working feature to the project owner and core collaborators.
- Implemented the sync model on Socket.IO: live presence with per-user color coding, per-node soft locks with “Editing: name” badges, accept/decline code-change proposals with first-accept-wins conflict resolution, and execution-output propagation so a node run on one browser surfaces results on every collaborator’s downstream nodes.
- Handled edge and node deletion conflicts with role-aware Accept/Decline cards; gated all collaboration behavior behind an opt-in `--collab` flag so default single-user mode was untouched.
- Maintainers adopted the prototype into a `collaboration-revised` feature branch and rewrote the implementation against newer architecture; event vocabulary and proposal/lock model from the prototype ship in the released feature, with credit by name in the project’s official documentation at `urban-toolkit/curio/docs/COLLABORATION.md`.